Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Multi Observation Production Centre

Global Ocean Multi Observation OMEGA3D product MULTIOBS_GLO_PHY_ UVW_3D_MYNRT _015_007

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	December 2019	All	Initial version of the document	B. Buongiorno Nardelli	Mercator Ocean
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3.0	May 2025	All	Updated content to NOV25 release	B. Buongiorno Nardelli, S. Asdar, L. Della Cioppa	H.Etienne

ACRONYM TABLE

Acronym	Description
TAC	Copernicus Marine Service Thematic Assembly Centre
NRT	Near Real Time
REP	Reprocessing
MY	Multi-year
SLA	Sea Level Anomaly
SST	Sea Surface Temperature
SSS	Sea Surface Salinity
Argo	International profiling float network (www.argo.net)
XBT	Expendable Bathythermograph
CTD	Conductivity, Temperature, and Depth; It refers to a package of electronic instruments that measure these properties
EAN	Estimated Accuracy Number
EOF	Extended Orthogonal Function
MLD	Mixed Layer Depth

KEY DEFINITIONS

N/A

RELEVANT DOCUMENT LIST

N/A

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the product **MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007**[1], which provides observation-based 3D quasi-geostrophic (QG) vertical and horizontal ocean currents from the surface to 1500 m depth over a regular grid at ¼° horizontal resolution, both in Near Real Time (NRT) and as Multi Year (MY) datasets. The datasets delivered are:

- cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i, which corresponds to NRT (previous week) u, v, w weekly sampled fields (representative of each Wednesday)
- cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i, which corresponds to MY u, v, w weekly sampled fields (representative of each Wednesday) over the period January 1993 to year-1.

The MY dataset delivers velocities obtained by solving a Q-vector formulation of the Omega equation [DR1] applied to previous (weekly) version of ARMOR3D data (MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012), while the NRT dataset is based on the new daily NRT ARMOR3D dataset delivered at 1/8° resolution (downgraded to ¼° weekly resolution).

I.2 Summary of the results

The accuracy of the ocean current velocity components depends on both the accuracy of the input data and on the theoretical limits of the diagnostic models and parameterizations used. The quasi-geostrophic (QG) Omega solved here cannot be applied at low latitudes, and several assumptions are made to parameterize vertical mixing, including the use of empirical models of the currents in the Ekman layer. An increase of the error in the equatorial zone and in coastal areas is thus expected, and the product is basically suited only for open ocean applications.

Direct assessment of the vertical component accuracy is not possible due to the lack of direct reference observations. Typical values of retrieved vertical velocities attain around O(10-100 m/day). Vertical velocities resulting from the diabatic forcing alone closely resemble the vertical velocities that can be obtained from classical Ekman pumping/suction at the base of the Ekman layer, both in terms of intensity and main patterns.

The horizontal ageostrophic components are estimated once the QG vertical velocities are known (see Section II.2.2). Total horizontal and geostrophic components, obtained as the difference between the total and the ageostrophic components, can thus be compared with independent estimates of the ocean currents retrieved from drifting buoys and Argo floats displacements [DR 3, DR 4].

For the MY dataset, the global values of the Root Mean Square Differences (RMSD) of the total horizontal velocities attain around 0.133 m/s at the surface, 0.105 m/s at 15 m and 0.057 m/s at 1000 m

[1] *This product supersedes former MULTIOBS_GLO_PHY_UVW_3D_REP_015_007.*

depth, with significant geographical variations. Overall, surface values (at 0 and 15 m) display a slightly negative bias while values at 1000 m are slightly overestimated.

For the NRT dataset, the global values of the Root Mean Square Differences (RMSD) of the total horizontal velocities attain around 0.106 m/s at the surface and 0.118 m/s at 15 m, with significant geographical variations. Overall, surface values (at 0 and 15 m) display a (slightly) negative bias.

I.3 Estimated Accuracy Numbers

Global accuracy numbers for MULTIOBS_GLO_PHY_UVW_3D_REP_015_007 are computed by comparison of total velocities with fully independent velocity estimations from undrogued/drogued drifters at 0 m and 15 m depth [DR 3], respectively, and Argo float derived velocity at 1000 m [DR 4]. These numbers (Table 1) are not representative of effective accuracy at local/regional scales, which can only be described by spatial distributions of the metrics (Section IV).

Table 1: Estimated accuracy numbers. All values have been truncated at the statistically significant decimal place (calculating 95% confidence intervals through a Monte Carlo approach). All metrics (defined in section III) have been estimated for both zonal and meridional velocity components (U, V) and velocity magnitude ($|\vec{u}|$).

Dataset: cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i			
Variable	**Metrics**	**Value (m/s)**	**# matchups**
U **(0 m)**	**bias**	0.004	870603
V **(0 m)**	**bias**	0.001	870603
$\|\vec{u}\|$ **(0 m)**	**bias**	-0.067	870603
U **(15 m)**	**bias**	0.004	588387
V **(15 m)**	**bias**	-0.001	588387
$\|\vec{u}\|$ **(15 m)**	**bias**	-0.036	588387
U **(1000 m)**	**bias**	0.001	1245053
V **(1000 m)**	**bias**	0.000	1245053
$\|\vec{u}\|$ **(1000 m)**	**bias**	0.015	1245053
U **(0 m)**	**RMSD**	0.136	870603
V **(0 m)**	**RMSD**	0.125	870603
$\|\vec{u}\|$ **(0 m)**	**RMSD**	0.133	870603
U **(15 m)**	**RMSD**	0.105	588387
V **(15 m)**	**RMSD**	0.101	588387
$\|\vec{u}\|$ **(15 m)**	**RMSD**	0.103	588387
U **(1000 m)**	**RMSD**	0.067	1245053
V **(1000 m)**	**RMSD**	0.063	1245053
$\|\vec{u}\|$ **(1000 m)**	**RMSD**	0.057	1245053

Table 1: (continued)

Dataset: cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i			
Variable	**Metrics**	**Value (m/s)**	**# matchups**
U **(0 m)**	**bias**	-0.002	50148
V **(0 m)**	**bias**	0.001	50148
$\|\vec{u}\|$ **(0 m)**	**bias**	-0.032	50148
U **(15 m)**	**bias**	0.007	50008
V **(15 m)**	**bias**	0.003	50008
$\|\vec{u}\|$ **(15 m)**	**bias**	-0.052	50008
U **(1000 m)**	**bias**	-	0
V **(1000 m)**	**bias**	-	0
$\|\vec{u}\|$ **(1000 m)**	**bias**	-	0
U **(0 m)**	**RMSD**	0.107	50148
V **(0 m)**	**RMSD**	0.104	50148
$\|\vec{u}\|$ **(0 m)**	**RMSD**	0.106	50148
U **(15 m)**	**RMSD**	0.129	50008
V **(15 m)**	**RMSD**	0.112	50008
$\|\vec{u}\|$ **(15 m)**	**RMSD**	0.118	50008
U **(1000 m)**	**RMSD**	-	0
V **(1000 m)**	**RMSD**	-	0
$\|\vec{u}\|$ **(1000 m)**	**RMSD**	-	0

II PRODUCTION SYSTEM DESCRIPTION

SHORT DESCRIPTION

Production centres names: MULTIOBS-CNR-ROMA-IT

Production centre description: This product has been developed and delivered by the Consiglio Nazionale delle Ricerche, Istituto di Scienze Marine (in English, Italian National Research Council, Institute of Marine Sciences), that is one of the contributors to the Copernicus Marine Service MOB-TAC (MultiOBservation Thematic Assembly Centre).

Production system name: Global Ocean Multi Observations OMEGA3D Product (CMEMS name: MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007)

- **Datasets:** cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i
- **Datasets:** cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i

Description / production system name: OMEGA3D-MYNRT

II.1 Input datasets

Three input data sets are used to compute the MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007 product:

- the global ARMOR3D reprocessed product [DR 8, DR 9] distributed within CMEMS MOB-TAC as MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 - old dataset-armor-3d-rep-weekly and new cmems_obs-mob_glo_phy_nrt_0.125deg_P1D-m, which provide 3D temperature (T), salinity (S), and absolute height (H);
- the multi-year L4 of global (Ekman) total velocity field at 0 m and 15 m, including zonal and meridional velocity at a 1h frequency over 1/4° regular grid (distributed within CMEMS MOB-TAC as product ID: MULTIOBS_GLO_PHY_MYNRT_015_003) [DR 10];
- the surface fluxes provided by ECMWF (European Centre for Medium-Range Weather Forecasts) global atmospheric reanalysis ERA-5 [DR 2].

II.2 OMEGA3D processing

The 3D fields of quasi-geostrophic ocean currents, including vertical velocities, are retrieved here through a diagnostic tool (hereafter called OMEGA3D) based on a diabatic Q-vector formulation of the Omega equation [DR 1]. OMEGA3D is applied to the fields of temperature, salinity and geostrophic currents provided by MOB-TAC ARMOR3D MY product, and diabatic forcing terms are estimated through a combination of an empirical reconstruction of Ekman currents and K-Profile Parameterization (KPP) combining ARMOR3D data, MOB-TAC Copernicus/Globcurrent data and ERA5 surface fluxes.

ARMOR3D combines satellite and in situ observations and carries out its 3D reconstruction first through a multi-linear regression between surface data and deep values, and then through an optimal interpolation of in-situ T/S profiles [DR 8, DR 9]. ARMOR3D data are reconstructed on a regular grid with

50 levels, regularly spaced at different intervals depending on the depth ranges between the surface and 1500 m depth. The old dataset-armor-3d-rep-weekly (used for MY processing) provides one daily 3D reconstruction per week at 1/4° resolution, while the new cmems_obs-mob_glo_phy_nrt_0.125deg_P1D-m provides daily data at 1/8° resolution. The 1/8° fields are preliminarily remapped onto OMEGA3D grid before estimating all terms needed in the equation.

To better describe upper layer dynamics, OMEGA3D runs on a non-uniform vertical grid, with a refined mesh close to the upper boundary, and only uses a second-order accuracy finite difference scheme to estimate derivatives [DR 11]. Input data are thus first interpolated through a spline on 78 vertical levels with layer thickness increasing as the square of depth (10 of which are located in the first 20 m, out of the 20 layers found in the first 100 m). The final grid includes 75 vertical levels between 2.5 m and 1482.5 m. Before running the OMEGA3D, potential density profiles are also adjusted to impose static stability [DR 1].

ERA5 is a global reanalysis for the atmosphere, ocean and land surface that provides outputs of a range of variables with temporal and spatial continuity over the period 1979 to present. A description of the ERA5 1979-present system is provided in [DR 2, DR6]. ERA5 data processing is carried out using ECMWF Earth System model IFS, cycle 41r2 (see IFS Documentation CY41R2)[2]. IFS assimilates several observations of upper-air atmospheric variables (e.g. satellite radiances, temperature, wind vectors, specific humidity, and ozone). For this product, we have extracted and mapped on our grid the daily mean fields of the zonal and meridional components of the turbulent surface stress, the surface latent and heat fluxes, the surface net solar and thermal radiation, as well as total precipitation and evaporation (which were both used to estimate the equivalent surface salinity flux).

II.2.1 Omega equation

Different formulations of the Omega equation can be found in the literature, depending on the approximation considered and on the analytical steps followed to derive it [DR 1]. Here, we consider the following diabatic quasi-geostrophic (QG) Q-vector formulation of Omega equation:

$$\nabla^2{}_h(N^2 w) + f^2 \frac{\partial^2 w}{\partial z^2} = \nabla_h \cdot \boldsymbol{Q} \tag{1}$$

In Eq. (1), w represents the vertical velocity (positive upwards), N^2 is the Brunt–Väisälä frequency, f is the Coriolis parameter, h indicates the horizontal components, z is depth, and the $\boldsymbol{Q}$ vector includes three components reflecting different processes (kinematic deformation, twg, turbulent buoyancy, th, and turbulent momentum, dm), as defined below:

$$\boldsymbol{Q} = 2\boldsymbol{Q}_{twg} + \boldsymbol{Q}_{th} + \boldsymbol{Q}_{dm} \tag{2}$$

where

[2] https://www.ecmwf.int/en/publications/ifs-documentation

$$\mathbf{Q_{twg}} = \frac{g}{\rho_0}\left(\frac{\partial u_g}{\partial x}\frac{\partial \rho}{\partial x} + \frac{\partial v_g}{\partial x}\frac{\partial \rho}{\partial y}, \frac{\partial u_g}{\partial y}\frac{\partial \rho}{\partial x} + \frac{\partial v_g}{\partial y}\frac{\partial \rho}{\partial y}\right)$$

$$\mathbf{Q_{dm_woc}} = \frac{f}{\rho_0}\left(\frac{\partial^2}{\partial z^2}\left[\rho K_m\left(\frac{\partial v_g}{\partial z} + \frac{\partial v_{Ekman}}{\partial z}\right)\right], -\frac{\partial^2}{\partial z^2}\left[\rho K_m\left(\frac{\partial u_g}{\partial z} + \frac{\partial u_{Ekman}}{\partial z}\right)\right]\right)$$

$$\mathbf{Q_{th}} = -\frac{g}{\rho_0}\nabla_h\left(\frac{\partial}{\partial z}\left[K_\rho\left(\frac{\partial \rho}{\partial z} - \gamma_\rho\right)\right]\right) = \nabla_h\left(\frac{\partial}{\partial z}\left[K_\rho\left(N^2 + \frac{g}{\rho_0}\gamma_\rho\right)\right]\right)$$

In the above definitions, ρ_0 indicates the potential density, g is the gravitational acceleration, (u_g, v_g) and (u_a, v_a) represent the geostrophic and ageostrophic horizontal velocities, while turbulent terms are defined through classical non-local effective gradient, γ_ρ, and viscosity/diffusivity, K_m/ K_ρ.

Once the equation is solved for *w*, horizontal ageostrophic components can be estimated by integrating two expressions that are obtained during the analytical derivation of the omega equation:

$$u_a(z) = \int_{REF}^{z}\frac{\partial u_a}{\partial z}dz = \frac{1}{f^2}\int_{REF}^{z}\left(\frac{\partial}{\partial x}(N^2 w) - \frac{f}{\rho_0}\frac{\partial}{\partial z}\left(\frac{\partial \tau_{yz}}{\partial z}\right) - 2\frac{g}{\rho_0}\left(\frac{\partial u_g}{\partial x}\frac{\partial \rho}{\partial x} + \frac{\partial v_g}{\partial x}\frac{\partial \rho}{\partial y}\right) - \frac{g}{\rho_0}\frac{\partial}{\partial x}\left(\frac{\partial F_{\rho z}}{\partial z}\right)\right)dz \tag{3}$$

$$v_a(z) = \int_{REF}^{z}\frac{\partial v_a}{\partial z}dz = \frac{1}{f^2}\int_{REF}^{z}\left(\frac{\partial}{\partial y}(N^2 w) + \frac{f}{\rho_0}\frac{\partial}{\partial z}\left(\frac{\partial \tau_{xz}}{\partial z}\right) - 2\frac{g}{\rho_0}\left(\frac{\partial u_g}{\partial y}\frac{\partial \rho}{\partial x} + \frac{\partial v_g}{\partial y}\frac{\partial \rho}{\partial y}\right) - \frac{g}{\rho_0}\frac{\partial}{\partial y}\left(\frac{\partial F_{\rho z}}{\partial z}\right)\right)dz \tag{4}$$

where we have defined:

$$\tau_{xz} = \rho K_m\left(\frac{\partial u}{\partial z}\right)$$

$$\tau_{yz} = \rho K_m\left(\frac{\partial v}{\partial z}\right)$$

$$F_{\rho z} = K_\rho\left(\frac{\partial \rho}{\partial z} - \gamma_\rho\right)$$

and assumed that the ageostrophic velocities can be neglected at the reference layer considered in the integral (here taken as the deepest level).

II.2.2 Diabatic forcing term estimation

In the present version of the product, we adopted a parameterization of the diabatic forcing term based on the results of European Space Agency World Ocean Circulation project [DR 5]. Specifically, we take advantage of the empirical estimation of the horizontal Ekman velocity provided by MOB-TAC Copernicus/Globcurrent product [DR 10] to constrain the viscosity values.

Specifically, we assumed that the background Ekman velocity can be approximated through an analytical fit of the ageostrophic currents (provided at 0 m and 15 m by the empirical reconstruction) to a compressed Ekman spiral:

$$u_{Ekman}(z) = e^{\frac{z}{D_{amp}}}[\, u_0 \cos(z/D_{rot}) - \, v_0 \sin(z/D_{rot})] \tag{5}$$

$$v_{Ekman}(z) = e^{\frac{z}{D_{amp}}}[\, u_0 \sin(z/D_{rot}) + \, v_0 \cos(z/D_{rot})] \tag{6}$$

where (u_0,v_0) are the components of the empirical Ekman current at 0 m [DR 10] and D_{amp} *and* D_{rot} are the Ekman depth estimates obtained from the amplitude decay and vector rotation between 0 m and 15 m depth, respectively (see also [DR 7, DR 12, DR 13]).

The maximum viscosity, K_{max}, allowed within the Ekman layer was then derived from the local Ekman amplitude decay scale:

$$K_{max} = \frac{f D_{amp}^2}{2} \tag{7}$$

imposing an analytical profile within the boundary layer, chosen as in [DR 14]:

$$K_m(z) = K_{max}\left[1 + \tanh\left(\frac{z - D_{amp}}{\delta}\right)\right] \tag{8}$$

where δ represents the thickness of the transition layer (here set to 40 m, as in [DR 14]).

II.2.3 Omega solution

To solve Eq. (1), boundary conditions are needed. As in [DR 1], here we initially apply Dirichlet conditions at the surface and topographical boundaries, and Neumann conditions at the bottom and lateral boundaries of the domain. However, as the original grid is split in smaller horizontally overlapping sub-domains to reduce the computational effort, the inversion is carried out sequentially on these smaller sub-domains, imposing the vertical velocity values that resulted from the previous step as lateral boundary conditions for the subsequent calculations. Total vertical velocities are then obtained by summing the adiabatic and diabatic solutions and successively used to integrate Eq. (3) and (4) by a simple trapezoidal rule. An example of the vertical velocity fields is provided in fig. 1.

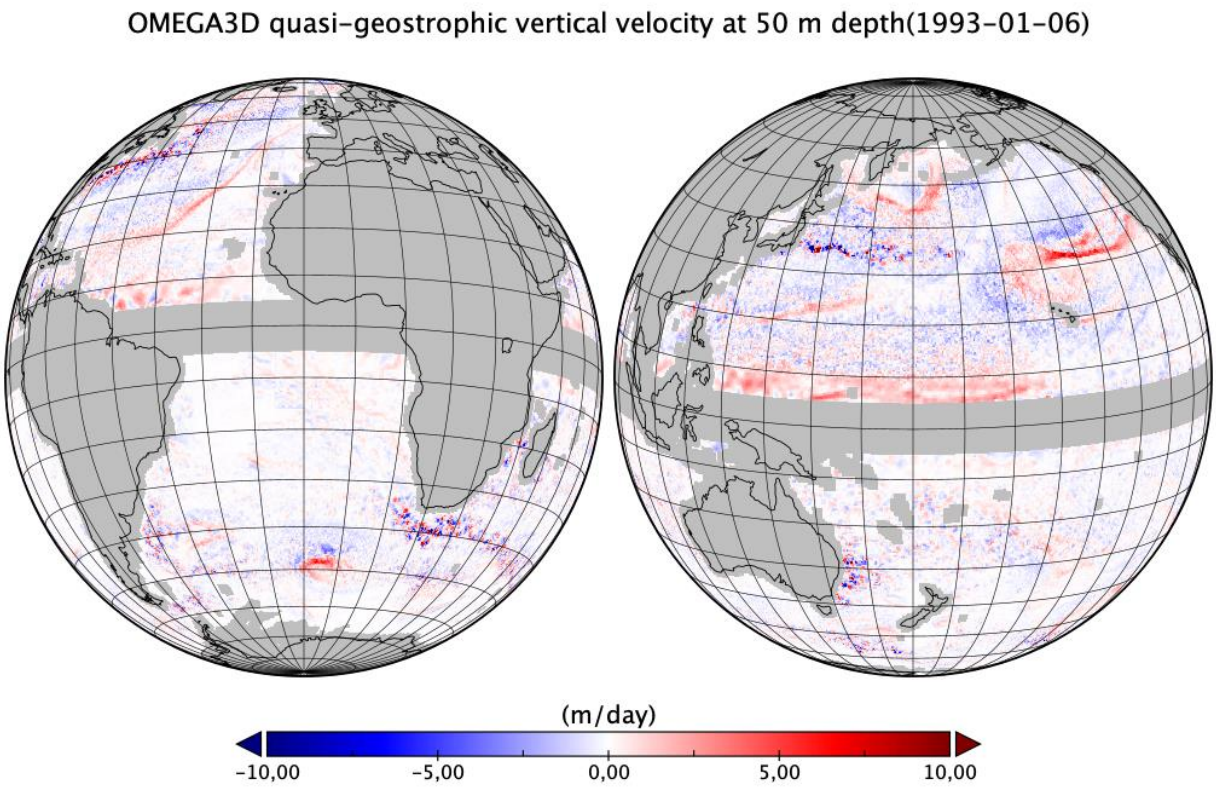

Figure 1: Example of cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-I vertical velocity field at 50 m depth (06/01/1993). Athis creas impacted by the topographical boundary conditions or limited by the validity of the QG approximation (i.e., close to the Equator) are masked in grey.

III VALIDATION FRAMEWORK

The OMEGA3D horizontal velocities were validated by comparison to Surface Velocity Program (SVP) data from NOAA Global Drifter Program (section IV.1) for both NRT and MY datasets and to YoMaHa'07 (Yoshinari Maximenko Hacker) database (section 0) for MY dataset only, focusing on the metrics listed in Table 2. In both cases, part of the root mean square differences, which attain around 0.13 m/s at the surface and 0.06 m/s at 1000 m, has to be attributed to the distinct information provided by velocities estimated from drifter displacements along a Lagrangian trajectory (including presence/absence of the drogue, averaging and filtering over space/ time) and by quasi-geostrophic Eulerian velocities retrieved through OMEGA3D.

Table 2: Metrics used for the quasi-geostrophic horizontal velocities' validation.

Name	Description	Parameter	Reference dataset	Period
UV-0m-W-CLASS4-BUOY-RMSD UV-0m-W-CLASS4-BUOY-BIAS UV-0m-W-CLASS4-BUOY-AVAIL	RMS and mean of the differences between SVP velocities and collocated fields	Zonal, meridional components and current intensity at 0 m	SVP velocities from NOAA Global Drifter Program [DR3]	1993/01 to 2024/06
.UV-15m-W-CLASS4-BUOY-RMSD UV-15m-W-CLASS4-BUOY-BIAS UV-15m-W-CLASS4-BUOY-AVAIL	RMS and mean of the differences between SVP velocities and collocated fields	Zonal, meridional components and current intensity at 15 m	SVP velocities from NOAA Global Drifter Program [DR3]	1993/01 to 2024/06
UV-1000m-W-CLASS4-BUOY-RMSD UV-1000m-W-CLASS4-BUOY-BIAS UV-1000m-W-CLASS4-BUOY-AVAIL	RMS and mean of the differences between YOMAHA velocities at 1000 m and collocated fields	Zonal, meridional components and current intensity at 1000 m depth	YoMaHa'07 dataset [DR]	1997/04 to 2018/12

A quantitative assessment of OMEGA3D vertical velocities cannot be carried out due to the lack of direct reference observations. However, a qualitative comparison between OMEGA3D diabatic component of the vertical velocities and classical estimates of the Ekman pumping/suction is presented in Section IV.3.

A matchup database was built starting from the undrogued/drogued SVP drifters 6-hourly data [DR 3], considering velocities measured within ±12 hours from the nominal reconstruction date (centred at 00:00 UTC). Individual SVP velocity data recorded within a grid pixel were compared with both the geostrophic flow and total velocities. Undrogued SVP derived velocities were compared to the first OMEGA3D layer velocities. Drogued SVP drift is deemed representative of currents at 15 m depth [DR 9]. As a consequence, SVP velocities were compared to linearly interpolated values between the fifth and the sixth levels (at 12.50 m and 16.25 m depth, respectively). The 6-hourly SVP data were low-pass filtered to remove inertial oscillations, by applying a moving average over a time window inversely scaled with the Coriolis parameter, as in [DR 1]. Maps of RMSD and BIAS have been obtained by considering matchups available within 5°x5° bins.

YoMaHa'07 database [DR 4] provides estimates of deep currents (at 1000 m) obtained from Argo float displacements at parking level (i.e., measured during each submerged phase of the profiler cycle). In fact, the Argo floats are designed to drift at a pre-selected parking pressure and reach the surface only to transmit data through ARGOS/IRIDIUM satellites, rising and submerging by modifying their buoyancy. Typically, a cycle (submerge-parking-emerge) lasts approximately 10 days. To increase the number of matchups, here YOMAHA data were taken within ±3 days from the nominal reconstruction date.

IV VALIDATION RESULTS

IV.1 Surface and 15 m current versus SVP drifters

The number of matchups between OMEGA3D datasets and SVP drifter observations are shown in Figure 2.

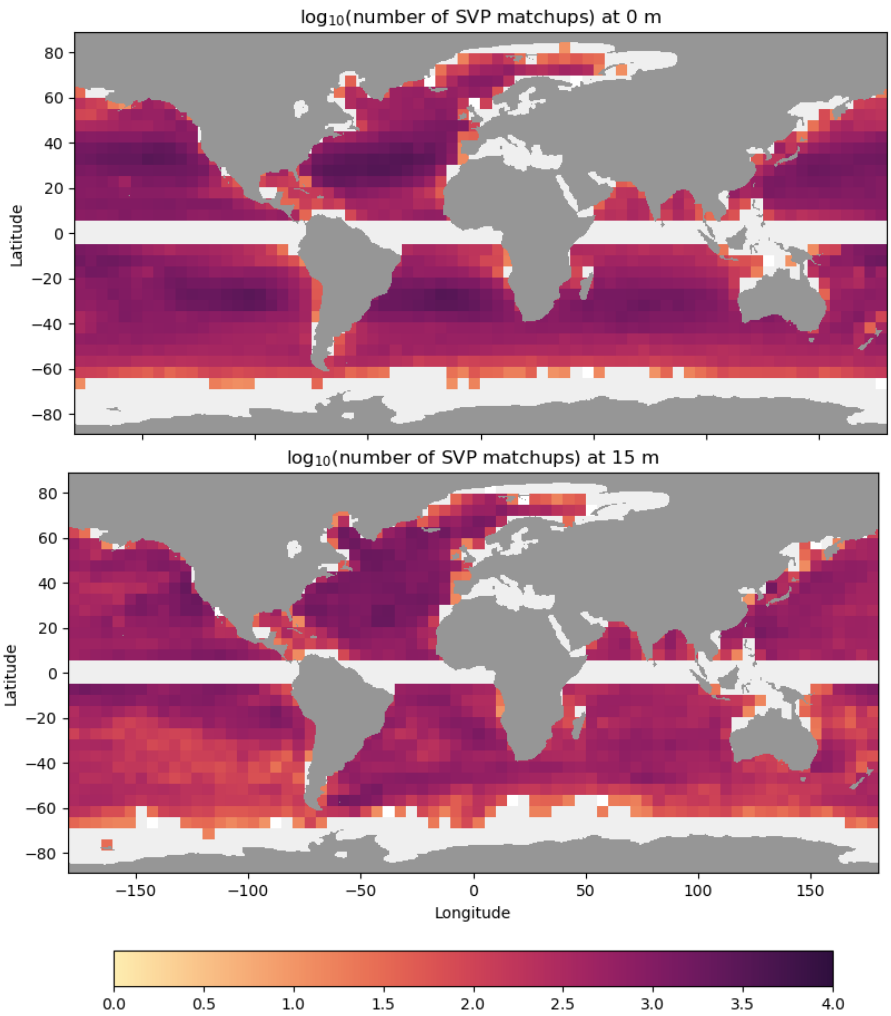

Figure 2a: UV-15m-W-CLASS4-BUOY-AVAIL: Number of 0m and 15 m MY dataset/SVP drifter matchups in 5° x 5° bins over the period 1993-2023. Common logarithmic scale.

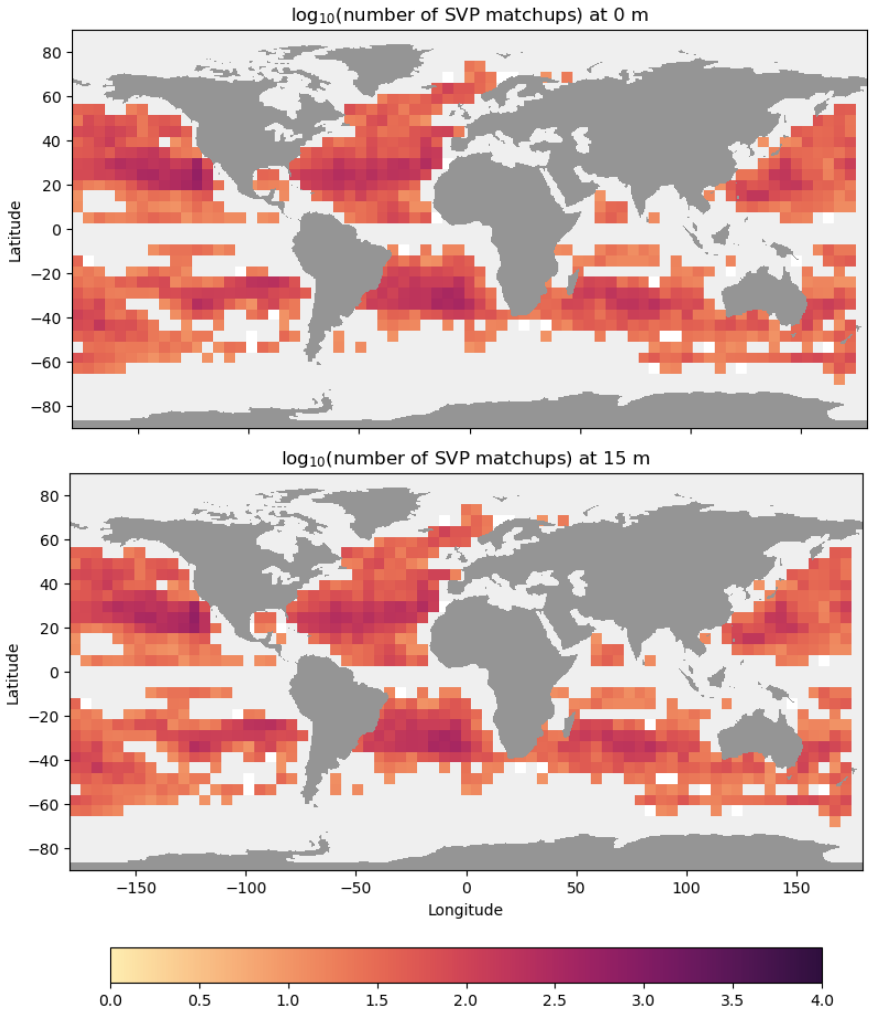

Figure 3b: UV-15m-W-CLASS4-BUOY-AVAIL: Number of om and 15 m NRT dataset/SVP drifter matchups in 5° x 5° bins over the period June 2023-June 2024. Common logarithmic scale.

Mean differences between SVP drifter velocities and OMEGA3D horizontal quasi-geostrophic currents reveal a general underestimation of the current intensities (Figure 4). MY Quasi-geostrophic velocities' bias is however reduced by a few cm/s (in absolute value) with respect to geostrophic estimates, mainly at low latitudes (Figure 6). The patterns of RMSD (Figure 8) are also very similar to what is found from our geostrophic computations based on altimetry. A reduction of a few cm/s is found comparing quasi-geostrophic velocities with geostrophic fields (Figure 8 and Figure 10) especially in low latitudes and along the Antarctic Circumpolar Current, both considering the MY and NRT datasets.

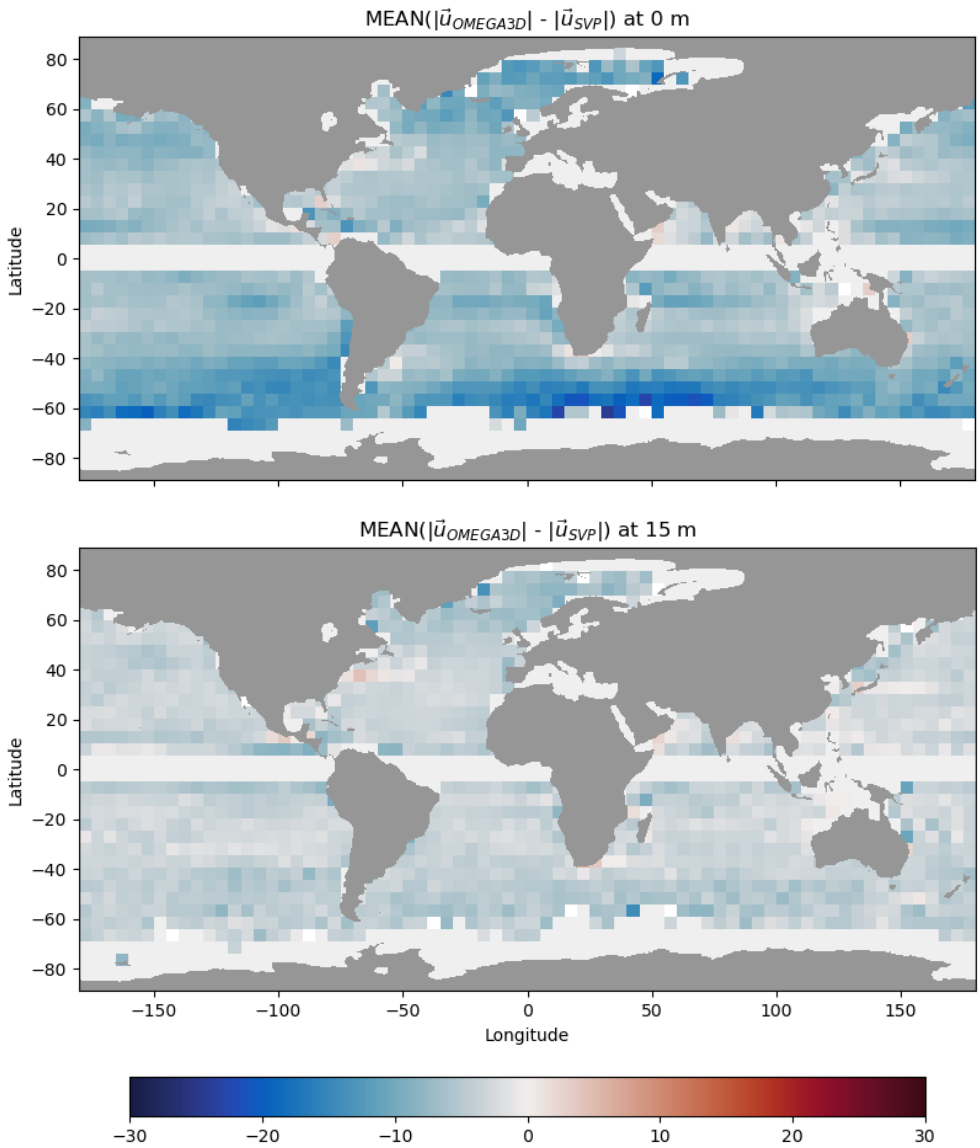

Figure 4a: UV-0m-W-CLASS4-BUOY-BIAS /UV-15m-W-CLASS4-BUOY-BIAS: Mean differences between SVP drifter velocities at 0 m (top) and 15 m (bottom) and MY OMEGA3D horizontal quasi-geostrophic currents intensity in 5° x 5° bins over the period 1993-2023. The units are given in 10^{-2} m/s.

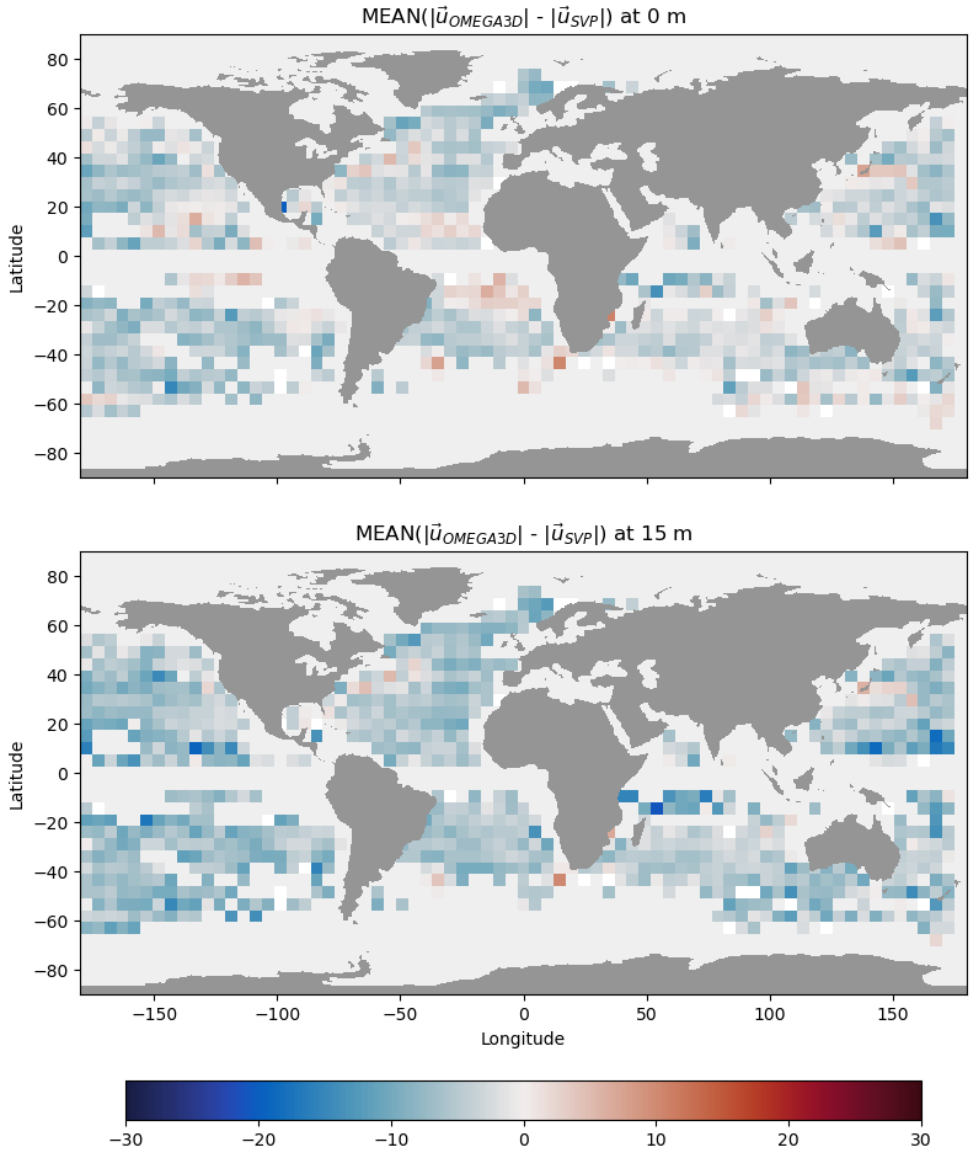

Figure 5b: UV-0m-W-CLASS4-BUOY-BIAS /UV-15m-W-CLASS4-BUOY-BIAS: Mean differences between SVP drifter velocities at 0 m (top) and 15 m (bottom) and NRT OMEGA3D horizontal quasi-geostrophic currents intensity in 5° x 5° bins over the period June 2023-June 2024. The units are given in 10^{-2} m/s.

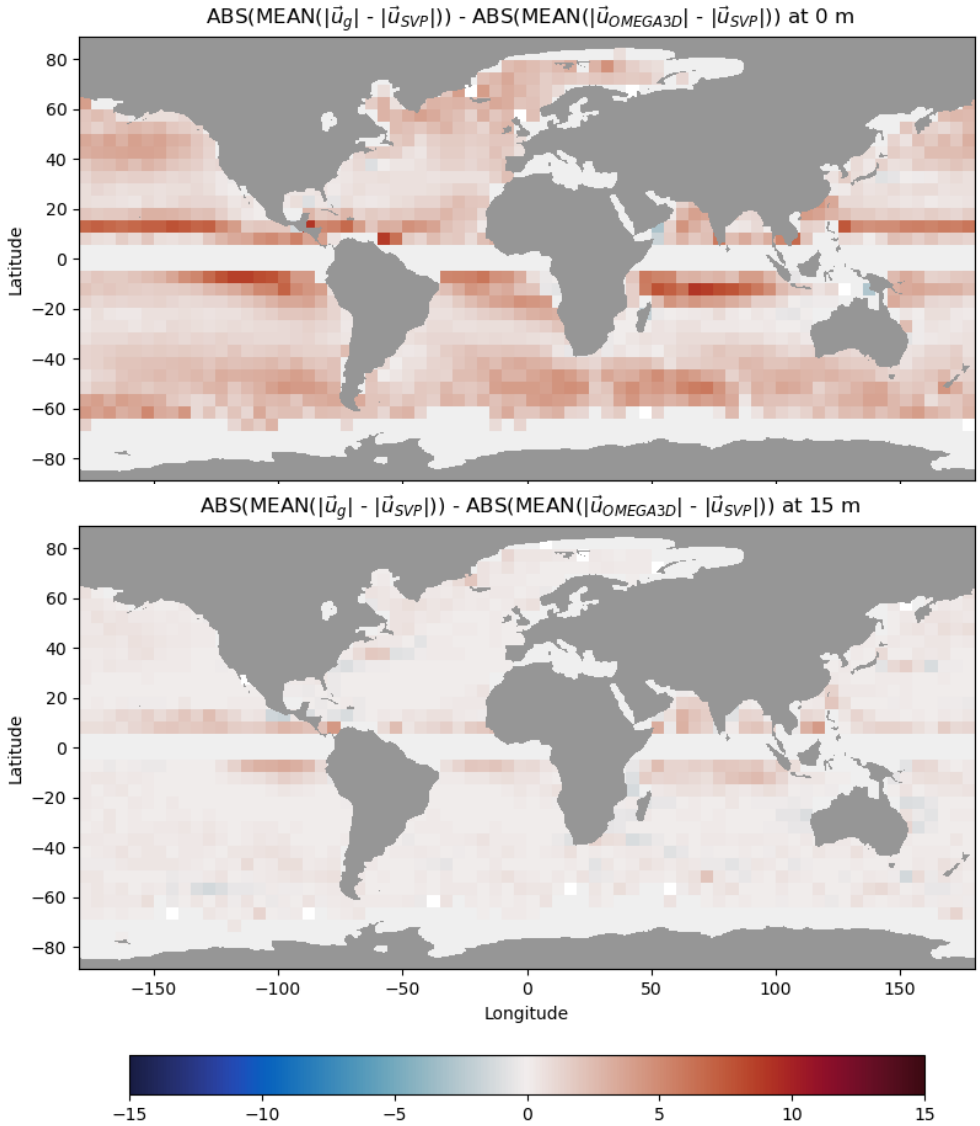

Figure 6a: Difference between absolute mean biases of MY OMEGA3D horizontal quasi-geostrophic and geostrophic velocities estimated vs SVP drifter velocities at 0 m (top) and 15 m (bottom) in 5° x 5° bins over the period 1993-2023. The units are given in 10^{-2} m/s. Positive values indicate an improvement with respect to geostrophy.

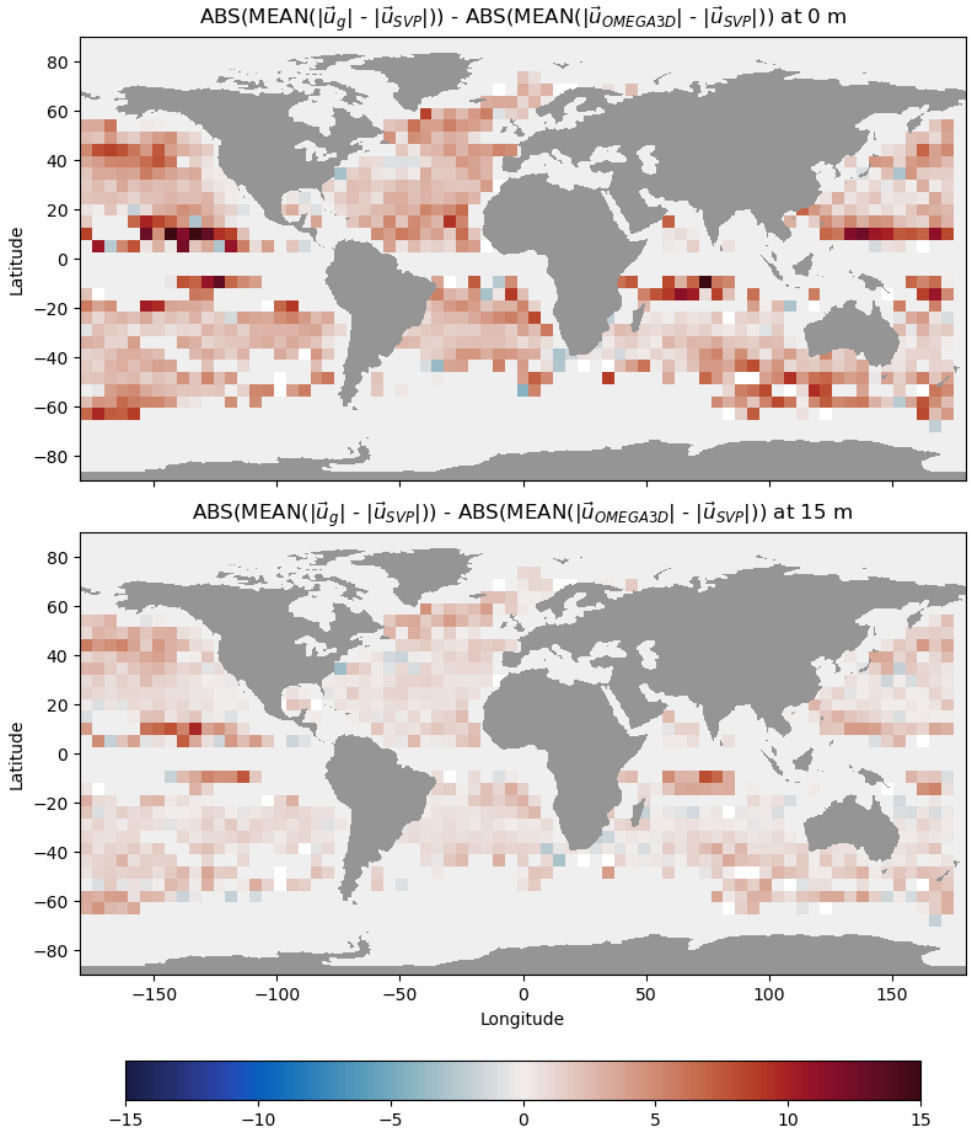

Figure 7b: Difference between absolute mean biases of NRT OMEGA3D horizontal quasi-geostrophic and geostrophic velocities estimated vs SVP drifter velocities at 0 m (top) and 15 m (bottom) in 5° x 5° bins over the period June 2023-June 2024. The units are given in 10^{-2} m/s. Positive values indicate an improvement with respect to geostrophy.

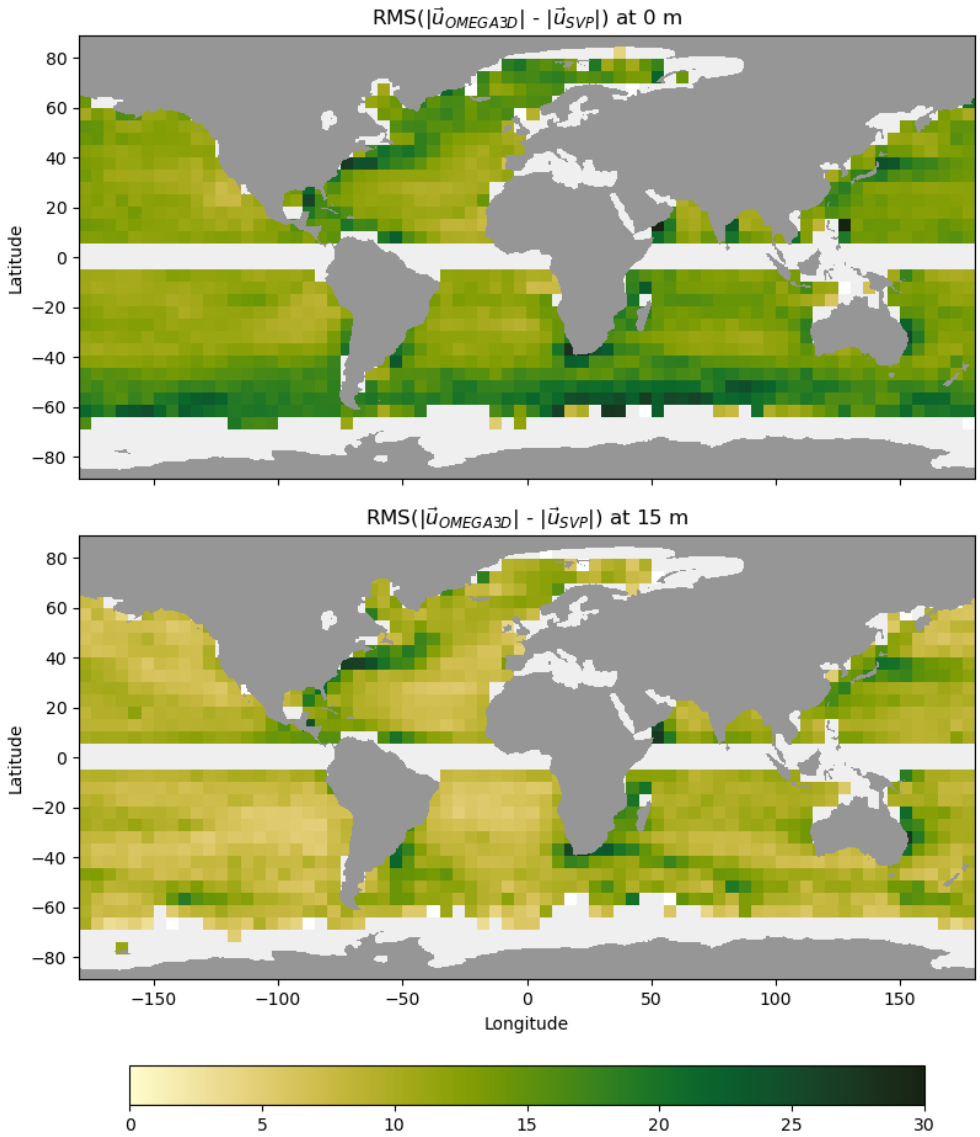

Figure 8a: UV-0m-W-CLASS4-BUOY-RMSD /UV-15m-W-CLASS4-BUOY-RMSD: Root mean squared differences (RMSD) between SVP drifter velocities at 0 m (top) and 15 m (bottom) and MY OMEGA3D horizontal quasi-geostrophic currents intensity in 5° x 5° bins over the period 1993-2023. The units are given in 10^{-2} m/s.

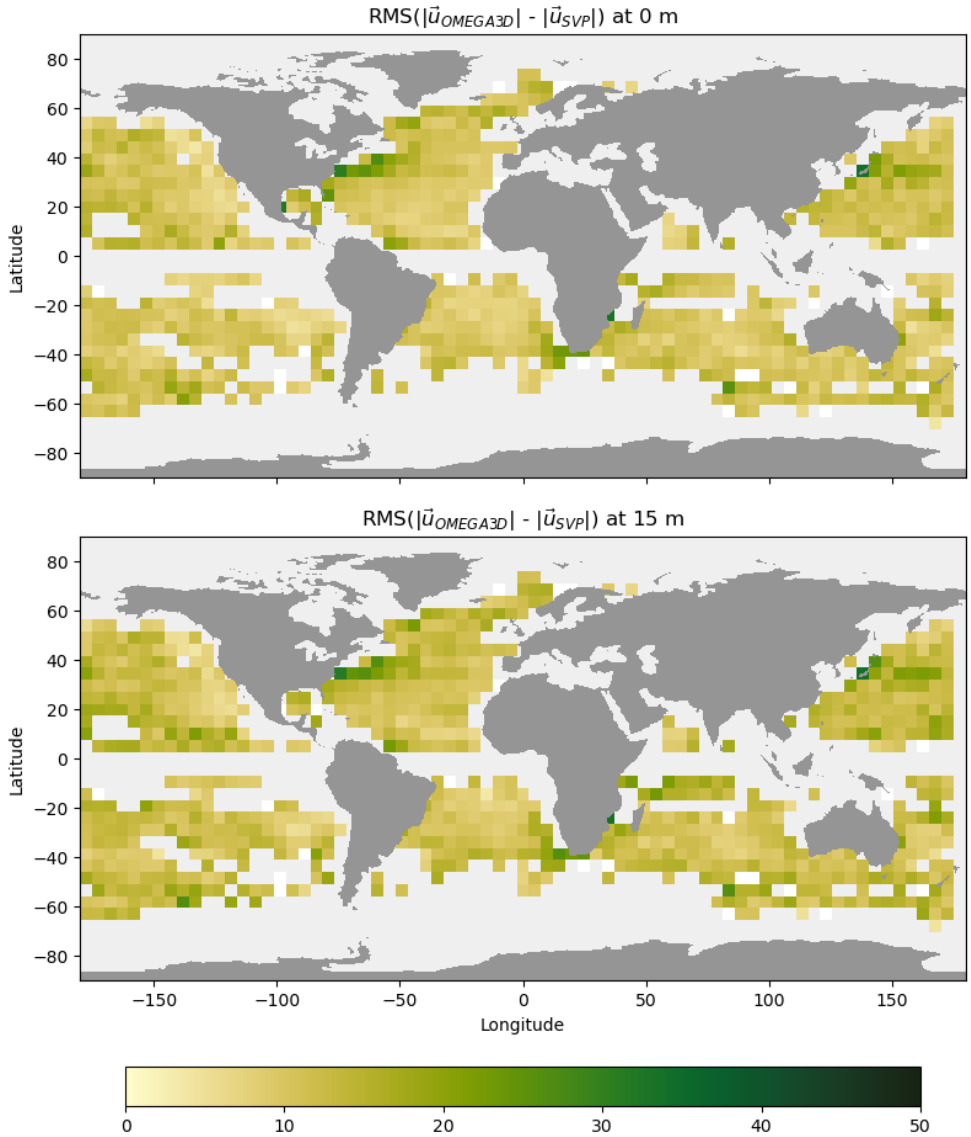

Figure 9b: UV-0m-W-CLASS4-BUOY-RMSD /UV-15m-W-CLASS4-BUOY-RMSD: Root mean squared differences (RMSD) between SVP drifter velocities at 0 m (top) and 15 m (bottom) and NRT OMEGA3D horizontal quasi-geostrophic currents intensity in 5° x 5° bins over the period June 2023-June 2024. The units are given in 10^{-2} m/s.

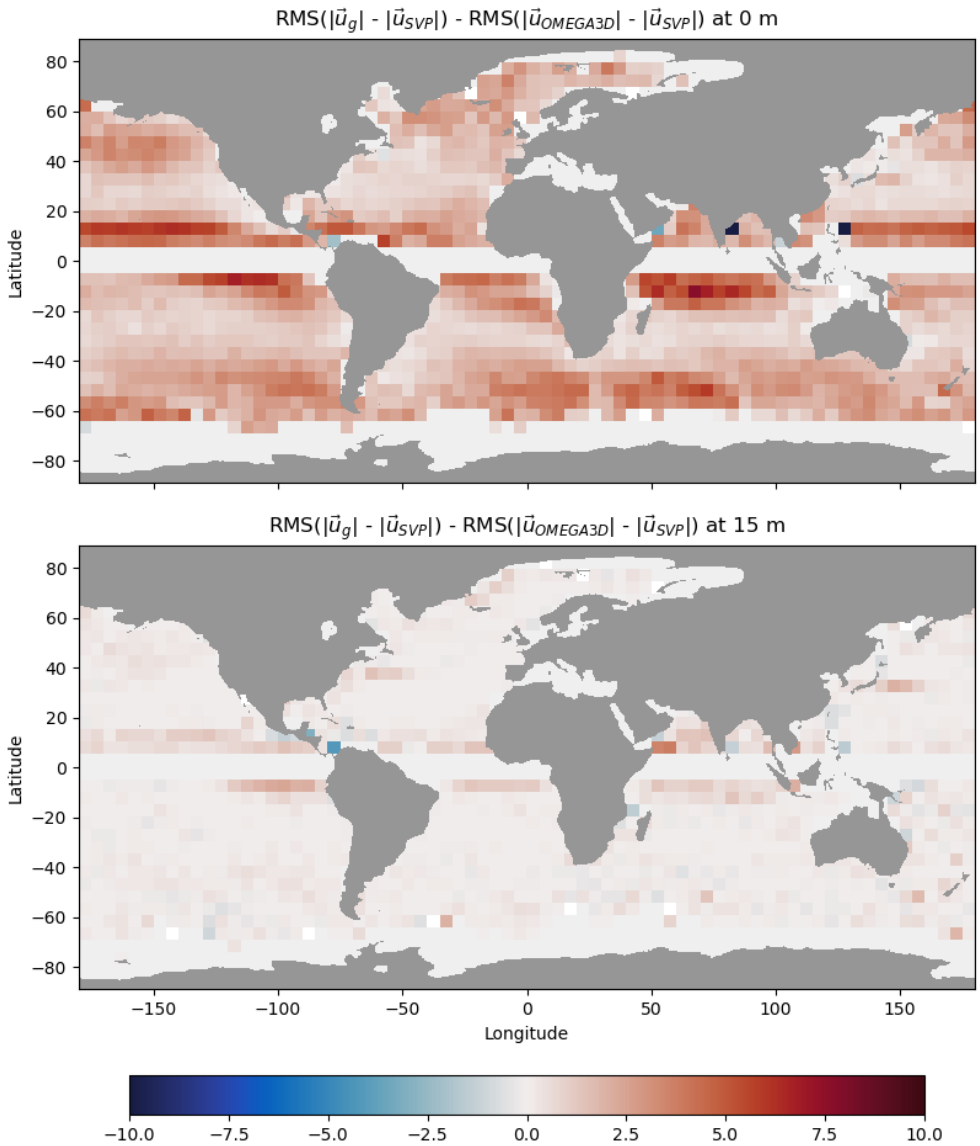

Figure 10a: Difference between Root mean squared differences (RMSD) of MY OMEGA3D horizontal quasi-geostrophic and geostrophic velocities estimated vs SVP drifter velocities at 0 m (top) and 15 m (bottom) in 5° x 5° bins over the period 1993-2023. The units are given in 10^{-2} m/s. Positive values indicate an improvement with respect to geostrophy.

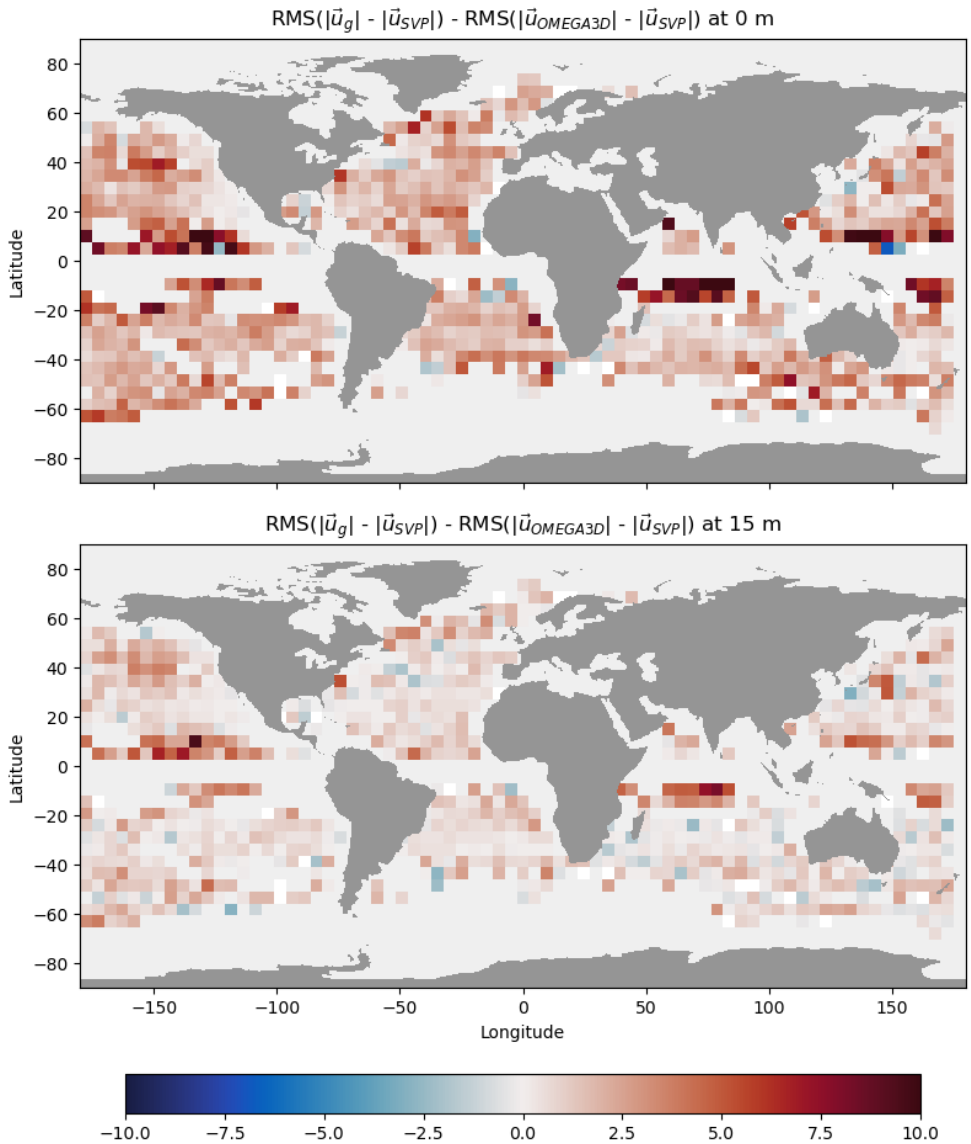

Figure 11b: Difference between Root mean squared differences (RMSD) of NRT OMEGA3D horizontal quasi-geostrophic and geostrophic velocities estimated vs SVP drifter velocities at 0 m (top) and 15 m (bottom) in 5° x 5° bins over the period June 2023-June 2024. The units are given in 10^{-2} m/s. Positive values indicate an improvement with respect to geostrophy.

IV.2 Deep quasi-geostrophic currents versus YoMaHa'07 currents at 1000 m

The comparison between OMEGA3D and YoMaHa'07 currents could be performed only for the MY dataset (Figure 7), because no data for comparison are available over the NRT validation period, see also Table 1.

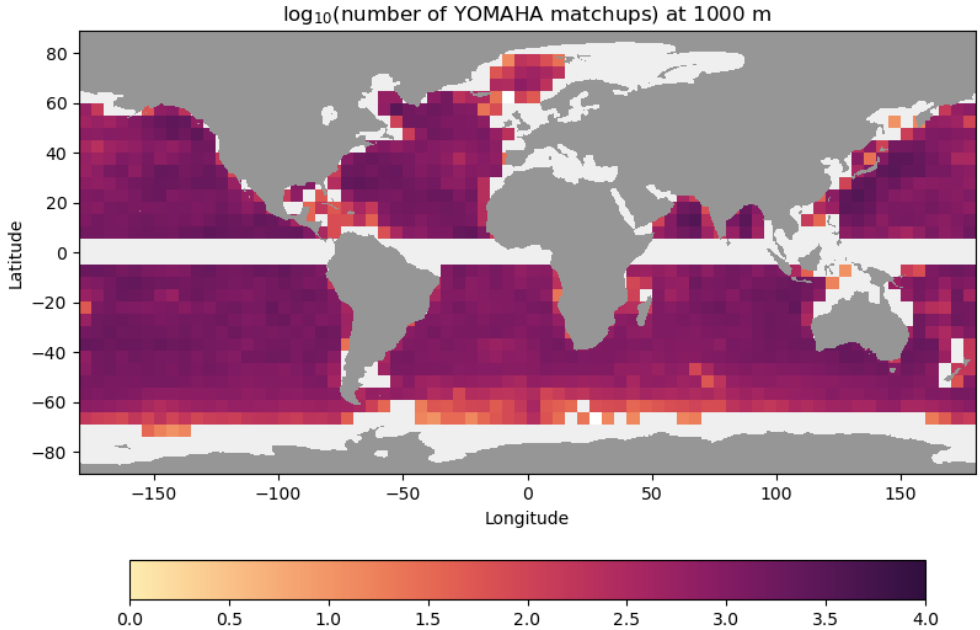

Figure 12: UV-1000m-W-CLASS4-BUOY-AVAIL Number of 1000 m YOMAHA matchups in 5° x 5° bins over the period 1993-2018. Common logarithmic scale.

Similarly to the results from the previous section, mean differences between YOMAHA velocities at 1000 m and corresponding OMEGA3D horizontal quasi-geostrophic currents basically present the same patterns that one would get from simple geostrophic velocities, with biases slightly increasing close to the equator, a general overestimation of the current intensities (Figure 13) and higher RMSD at frontal areas (Figure 14. Quasi-geostrophic horizontal velocity mean bias is reduced by a few mm/s (in absolute value, namely less than 10% of the actual error) with respect to deep geostrophic estimates in the main western boundary currents and along the Antarctic Circumpolar Current (Figure 15). A more marked improvement is found in the RMSD considering quasi-geostrophic current intensity with respect to geostrophic velocity, more noticeably in the Gulf Stream and Agulhas return current (Figure 16).

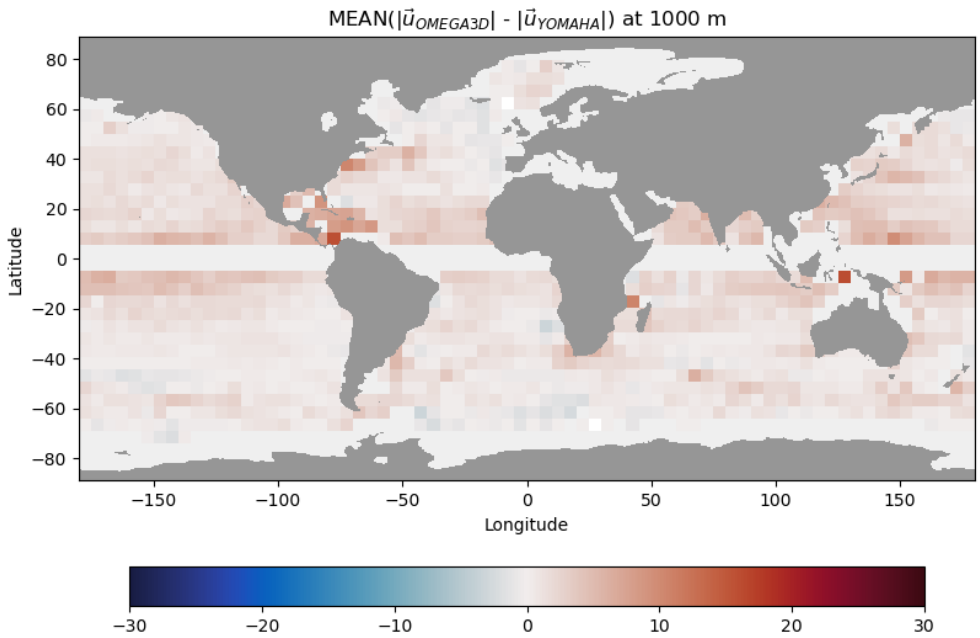

Figure 13: UV-1000m-W-CLASS4-BUOY-BIAS: Mean differences between YOMAHA velocities at 1000 m and MY OMEGA3D horizontal quasi-geostrophic currents intensity in 5° x 5° bins over the period 1993-2018. The units are given in 10^{-2} m/s.

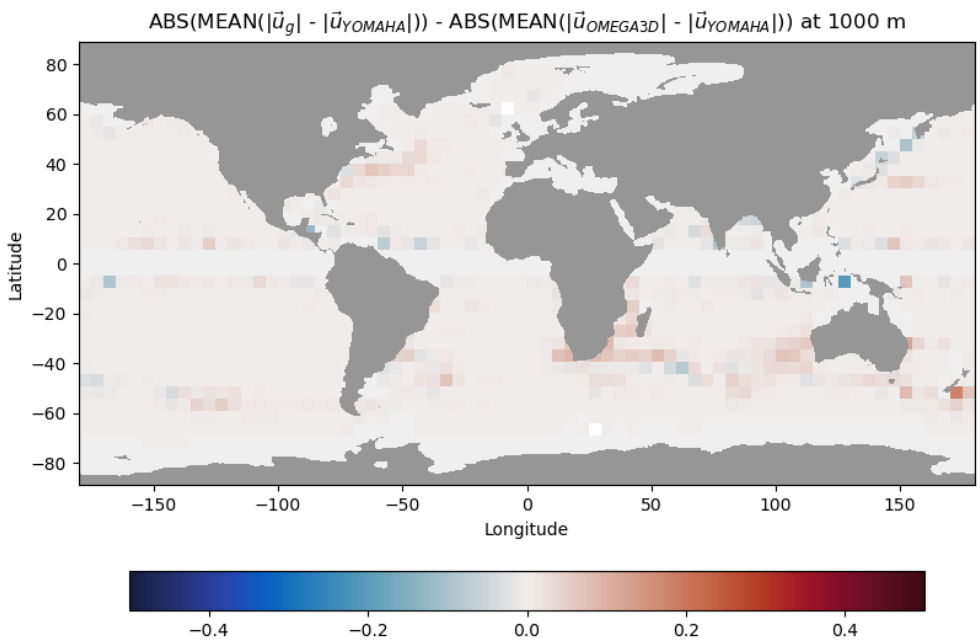

Figure 14: Difference between MBE of MY OMEGA3D horizontal quasi-geostrophic and geostrophic velocities estimated vs YOMAHA velocities at 1000 m in 5° x 5° bins over the period 1993-2018. The units are given in 10^{-2} m/s. Positive values indicate an improvement with respect to geostrophy.

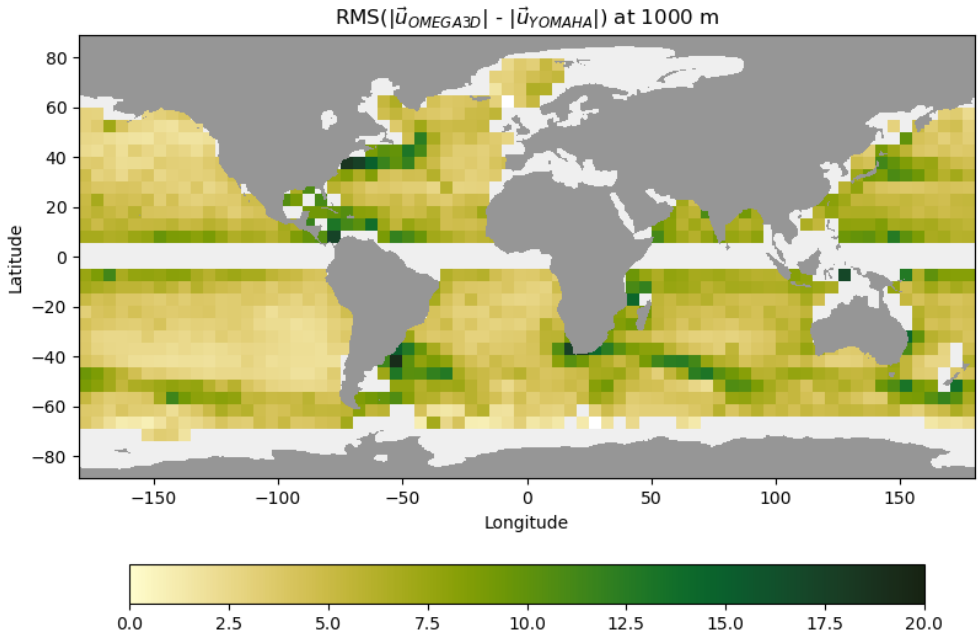

Figure 15: UV-1000m-W-CLASS4-BUOY-RMSD: Root mean squared differences (RMSD) between YOMAHA velocities at 1000 m and MY OMEGA3D horizontal quasi-geostrophic currents intensity in 5° x 5° bins over the period 1993-2018. The units are given in 10^{-2} m/s.

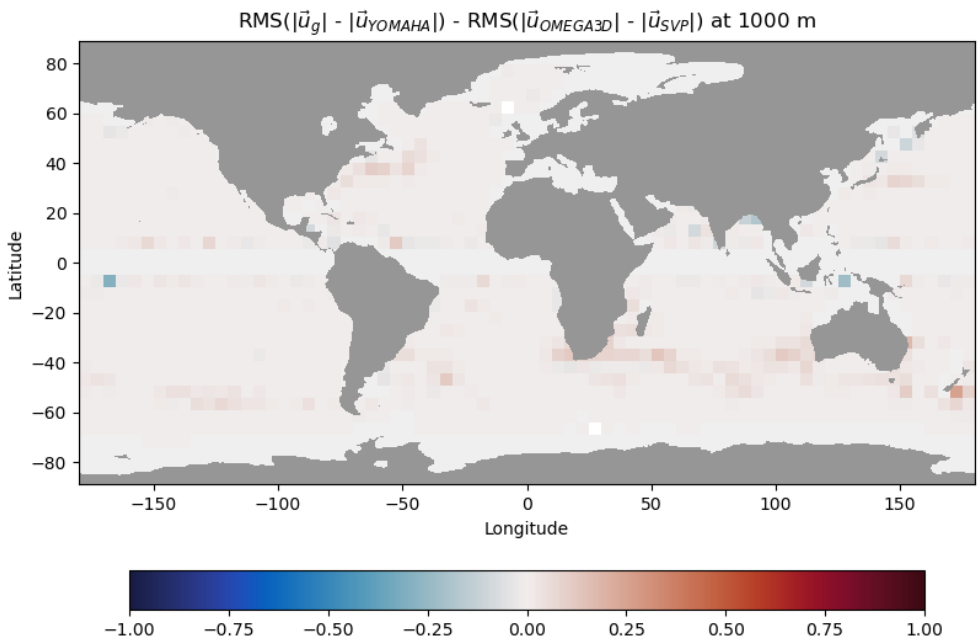

Figure 16: Difference between Root mean squared differences (RMSD) of MY OMEGA3D horizontal quasi-geostrophic and geostrophic velocities (estimated vs YOMAHA velocities at 1000 m), in 5° x 5° bin over the period 1993-2018 s. The units are given in 10^{-2} m/s. Positive values indicate an improvement with respect to geostrophy.

IV.3 Diabatic component of vertical velocities and Ekman pumping

We performed a comparison between MY OMEGA3D vertical velocity (Figure 17) and the estimates of the Ekman pumping/suction computed directly through its classical definition:

$$w_{Ek} = \frac{1}{\rho_0} \nabla \times \frac{\tau_{wind}}{f}, \tag{9}$$

as well as considering the Ekman pumping/suction modified by the surface geostrophic vorticity, following [DR 11]:

$$w_{Ek} = \frac{1}{\rho_0} \nabla \times \frac{\tau_{wind}}{f + \zeta}. \tag{10}$$

Here, we took the wind stress vector τ_{wind} from ERA data and estimated the relative vorticity ζ from OMEGA3D surface currents.

Ekman pumping/suction is an estimate of the vertical exchanges at the base of a theoretical layer (i.e. Ekman layer), which depends on eddy viscosity (assumed constant) and latitude (through the Coriolis parameter). Its value ranges between a few tens to a few hundreds of meters. In Figure 17 and Figure 18, the diabatic component and corresponding velocities computed from ERA data considering both the classical and the full (i.e. including also the effect of surface relative vorticity) Ekman models are compared for a sample day, respectively. Results show a strong agreement between the fields, both in terms of large-scale patterns and intensities, with OMEGA3D displaying more “noisy” patterns with respect to the Ekman velocities retrieved with Eq. (9), but slightly smoother patterns than Eq. (10) (this is likely due to the overestimation of the mean relative vorticity within the Ekman layer due to the use of surface values).

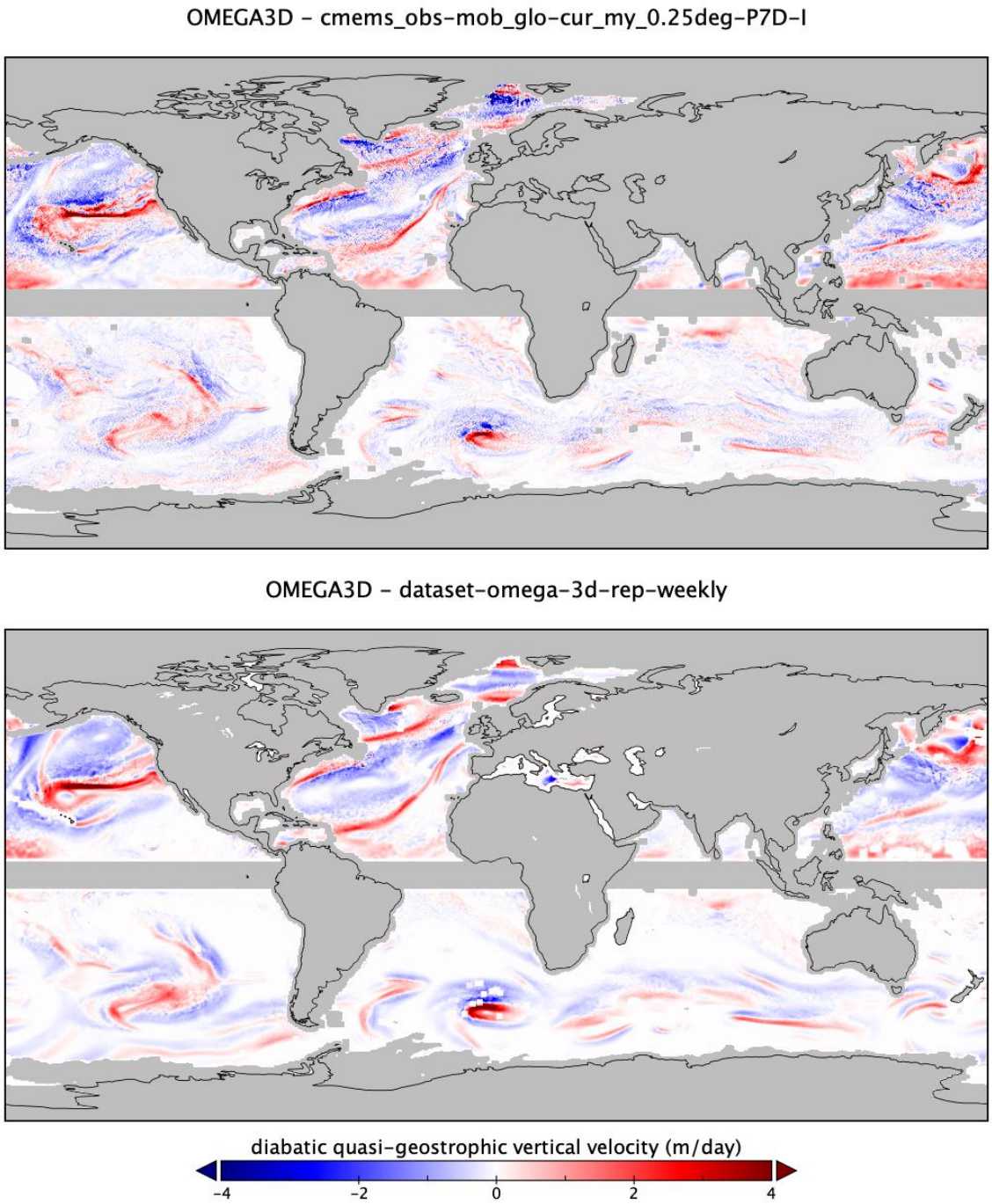

Figure 17: Diabatic component of MY OMEGA3D vertical velocity at 100 m depth (new version, cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-I, in the top panel, previous version, dataset-omega-3d-rep-weekly, in the bottom panel) on a sample day (1993-01-06). The units are given in m/day.

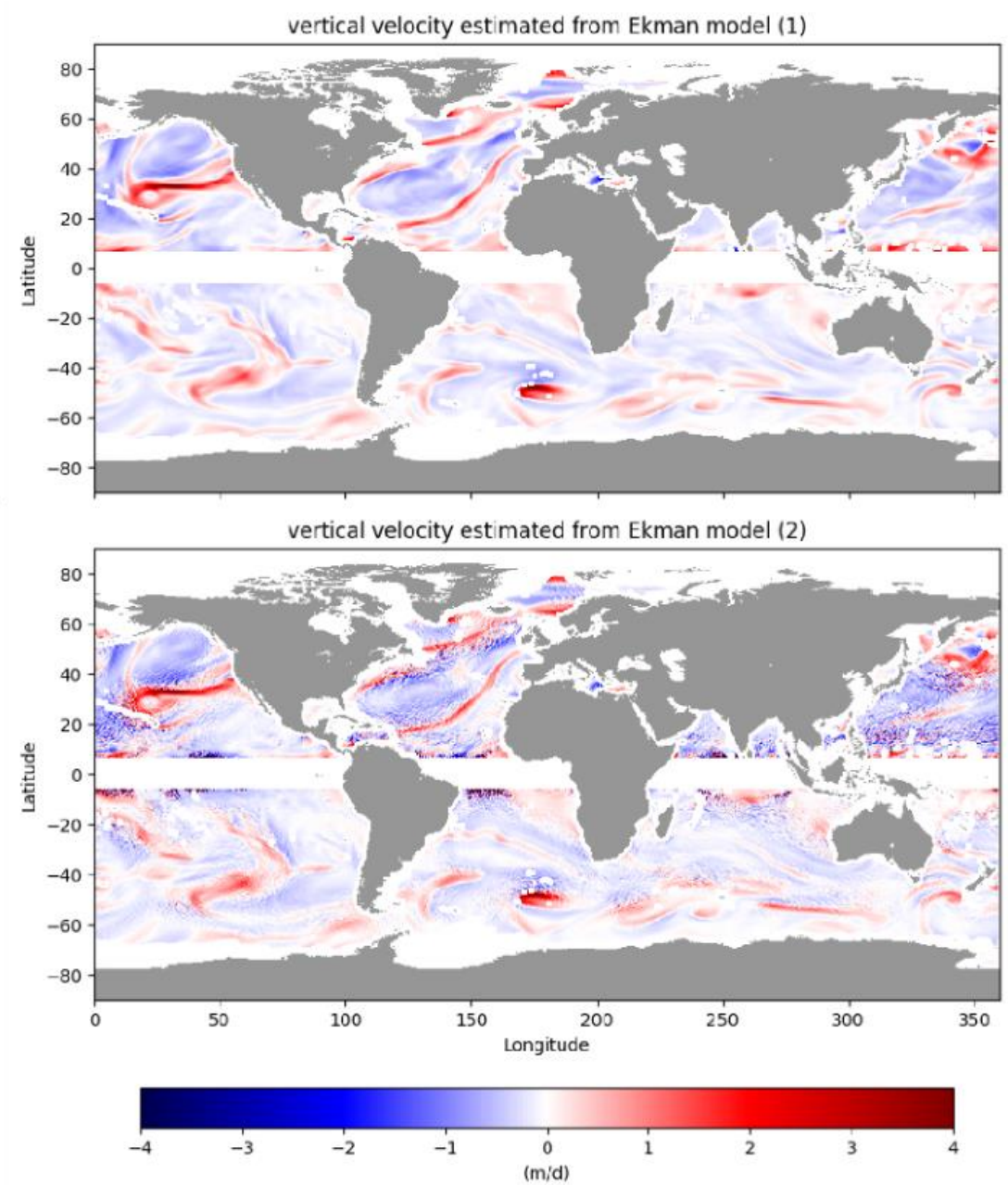

Figure 18: Classical Ekman velocities computed from ERA data on the same date as Figure 17 (top), and including also the effect of surface relative vorticity (bottom). The units are given in m/day.

V SYSTEM NOTICIABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system Including changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among others, which may impact product quality.

Date	**Description of changes and impacted product**
December 2019	**MULTIOBS_GLO_PHY_W_3D_REP_015_007** first release of *dataset-omega-3d-rep-weekly*
November 2024	**MULTIOBS_GLO_PHY_W_3D_REP_015_007** New dataset *cmems_obs-mob_glo-cur_my_0.25deg-P7D-i* supersedes former *dataset-omega-3d-rep-weekly*. The new version of the dataset provides an improved reconstruction of the diabatic components of the ageostrophic motion, by taking advantage of the modelled horizontal Ekman currents provided by product ID: MULTIOBS_GLO_PHY_MYNRT_015_003
November 2025	**Renamed product: MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007** New datasets: *cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i* *cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i*

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

MY dataset has not been changed. NRT dataset was not present in the previous version of the product.

VII REFERENCES

DR 1 **Buongiorno Nardelli, B.** A multi-year time series of observation-based 3D horizontal and vertical quasi-geostrophic global ocean currents. Earth Syst. Sci. Data 12, 1711–1723 (2020).

DR 2 **Hersbach H, Bell B, Berrisford P, Hurihara S, Horányi A et al. (2020).** The ERA5 global reanalysis.

DR 3 **Lumpkin, R., S. Grodsky, M.-H. Rio, L. Centurioni, J. Carton and D. Lee (2013),** Removing spurious low-frequency variability in surface drifter velocities. J. Atmos. Oceanic Techn., 30 (2), 353—360, http://dx.doi.org/10.1175/JTECH-D-12-00139.1.

DR 4 **Lebedev, K.V., H. Yoshinari, N. A. Maximenko, and P. W. Hacker (2007)**, YoMaHa'07: Velocity data assessed from trajectories of Argo floats at parking level and at the sea surface, IPRC Technical Note No. 4(2), June 12, 2007, 16p.

DR 5 **Asdar, S., Ciani, D. & Buongiorno Nardelli, B.** 3D reconstruction of horizontal and vertical quasi-geostrophic currents in the North Atlantic Ocean. Earth Syst. Sci. Data 16, 1029–1046 (2024).

DR 6 **Belmonte Rivas, M., and A. Stoffelen (2019),** Characterizing ERA-Interim and ERA5 surface wind biases using ASCAT, Ocean Sci., 15(3), 831–852, doi:10.5194/os-15-831-2019.

DR 7 **McWilliams, J. C., and E. Huckle (2006),** Ekman Layer Rectification, J. Phys. Oceanogr., 36(8), 1646–1659, doi:10.1175/jpo2912.1.

DR 8 **Guinehut, S., A.-L. Dhomps, G. Larnicol, and P.-Y. Le Traon (2012),** High resolution 3-D temperature and salinity fields derived from in situ and satellite observations, Ocean Sci., 8(5), 845–857, doi:10.5194/os-8-845-2012.

DR 9 **Mulet, S., M.-H. Rio, A. Mignot, S. Guinehut, and R. Morrow (2012),** A new estimate of the global 3D geostrophic ocean circulation based on satellite data and in-situ measurements, *Deep Sea Res. Part II Top. Stud. Oceanogr.*, *77–80*, 70–81, doi:10.1016/j.dsr2.2012.04.012.

DR 10 **Mulet, S., Rio, M.-H., Etienne, H., Artana, C., Cancet, M., Dibarboure, G., Feng, H., Husson, R., Picot, N., Provost, C., and Strub, P. T., (2021)**: The new CNES-CLS18 global mean dynamic topography, Ocean Sci., 17, 789-808.

DR 11 **Kalnay de Rivas, E. (1972)**, On the use of nonuniform grids in finite-difference equations, J. Comput. Phys., 10, 202–210.

DR 12 **Stern, M. (1965),** Interaction of a uniform wind stress with a geostrophic vortex. Deep-Sea Res. Oceanogr. Abstr., 12, 355–367, doi:10.1016/0011-7471(65)90007-0.

DR 13 **Roach, C. J.; Phillips, H. E.; Bindoff, N. L.; Rintoul, S. R.** Detecting and Characterizing Ekman Currents in the Southern Ocean. J. Phys. Oceanogr., 2015, 45 (5), 1205–1223. https://doi.org/10.1175/JPO-D-14-0115.1

DR 14 **Nagai, T.; Tandon, A.; Rudnick, D. L.** Two-Dimensional Ageostrophic Secondary Circulation at Ocean Fronts Due to Vertical Mixing and Large-Scale Deformation. J. Geophys. Res. 2006, 111 (C9), C09038. https://doi.org/10.1029/2005JC002964.